Reproduction of the nodal rotation patch test and the convergence test of Rajendran (2010) for $\theta \geq 90°$ with the USEM code base

USEM-MATLAB study `studies/rajendran_panel`

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Abstract

The square-panel tests of Rajendran [1, Secs. 5.5–5.7] (mid-side nodes rotated by $\theta = 90°$–$270°$, overlapping and folded elements) are recomputed with the spectral elements of the code base on gmsh meshes. With the 9-node element defined as in the paper (parametric test functions, metric trial functions in Cartesian coordinates, 9th node at the centroid of the corners, 5×5 Gauss points), all values of the 9-node Tables 15–19 are reproduced to four digits, for the unsymmetric (PM-QUAD9*) as well as the conventional (PP-QUAD9) and the metric (MM-QUAD9) elements, except for entries that are inconsistent within the paper itself (Section 6). The energy-norm convergence rates of PM-QUAD9* (Fig. 15 of the paper) are reproduced as well (2.00 and 1.98 against 2.11 and 1.97). For $p \geq 3$ the unsymmetric spectral element reproduces the cubic field of the shear test on every mesh, including fully folded ones, up to round-off. The hybrid trial space of the code base (PH: metric polynomials only for $P_p(\boldsymbol{x})$, parametric functions for the remainder) passes the constant and linear strain patch tests at $p = 2$ on all meshes, converges with rate 2.02 and 2.21 on sequences B and C, and reproduces the cubic field for $p \geq 3$; at $p = 8$ its round-off level on Meshes c and e is 100–1000 times lower than that of PM. The code base places interior nodes isoparametrically with the 8-node map; on these meshes this makes interior and edge nodes coincide for even p. The code base now stops with an error in this case and when the metric interpolation is singular, instead of returning a wrong (zero) solution (Section 5).

1 Set-up

Panel $[0, 2] \times [-1, 1]$, plane stress, $\nu = 0.25$, three point supports ($u_x = 0$ at $(0, \pm 1)$, $u_x = u_y = 0$ at $(0, 0)$), equilibrated loads on $x = 0$ and $x = 2$ (tension $\sigma_x = 1000$; moment $\sigma_x = -1500y$; shear $\sigma_x = -1500(2 - x)y$, $\tau_{xy} = 750(1 - y^2)$), displacement at $A = (2, 0)$. The meshes (Fig. 1) are generated with gmsh (`mesh/gen_panel_meshes.py`): 2×2 super-elements of 2×2 elements; the mid-side nodes of the internal edges of every super-element are rotated about its centre by θ (Meshes a–f: $\theta = 0°, 90°, 135°, 175°, 225°, 270°$; sequences A, B, C: $\theta = 0°, 90°, 135°$ with 1×1, 2×2, 4×4 super-elements). For $\theta \geq 90°$ every element of every mesh has $\det \boldsymbol{J} \leq 0$ somewhere.

Material constant. The paper states $E = 1500$, but its exact solutions (1.0, 1.5, 3.3125) are those of $E = 2000$ (elasticity solutions with the supports above: 1.333, 2.0, 4.4167 for $E = 1500$; 1.0, 1.5, 3.3125 for $E = 2000$). $E = 2000$ is used here; all displacements scale with $1/E$.

Loads. The consistent nodal forces (`matlab/rp_loads.m`) agree with the lists of the paper (Fig. 8 and Sec. 5.6) except for the moment case of Fig. 8, where the paper lists -250, -500 at nodes 8, 9 (and 250, 500 at nodes 14, 13); the consistent, antisymmetric values are -500, -250 (and 500, 250). The listed values would give a moment of 1125 instead of 1000; since the paper's PM results equal the exact 1.5, the consistent values were evidently used.

Elements. With $p = 2$ (9 nodes): PP9 = conventional element; MM9 = metric functions on both sides, $\int \boldsymbol{B}_R^{\mathsf{T}} \boldsymbol{D} \boldsymbol{B}_R \det \boldsymbol{J}$; PM9 = PM-QUAD9*, $\int (\boldsymbol{B}_L^*)^{\mathsf{T}} \boldsymbol{D} \boldsymbol{B}_R$. As in the paper: metric functions Q_2 in Cartesian coordinates relative to the element centroid (`ShapeFunctions2D_Metric`, method 0), 9th node at the centroid of the corners, 5×5 Gauss points. In addition: "PM, skew" = the code-base metric functions (skew coordinates, method 1) with the same nodes; "PM, code" = `Ke_2D` with the nodes of `pSpectralElement_NodesTopology` (4×4 points). For $p = 3, \ldots, 8$: PP and PM with interior nodes from the bilinear corner map (the generalisation of the corner centroid) and $2p \times 2p$ points, and $\mathsf{PM}_{\text{code}}$ with the interior nodes of the code base. The code-base metric functions are evaluated in the Lagrange basis on the bounding box of the skew coordinates (default); the trial space is that of the monomial basis. PH (hybrid trial space, `ShapeFunctions2D_Metric`, method 2):

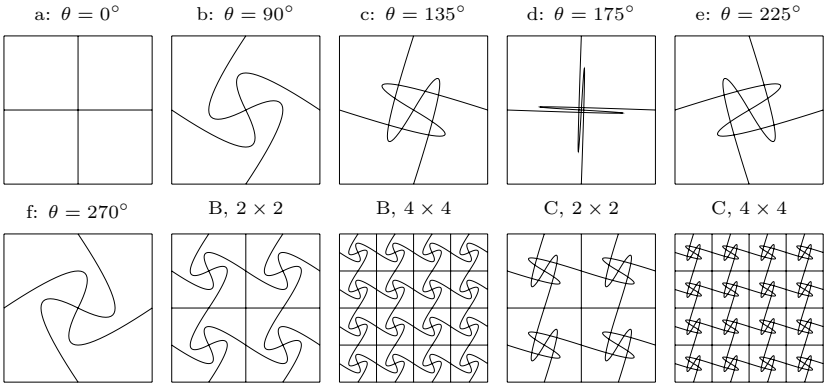


Figure 1: gmsh meshes: Meshes a–f of the patch test and the 2×2, 4×4 meshes of sequences B and C (cf. Figs. 9, 10 of the paper).

"PH, centr." with the 9th node at the corner centroid and 5×5 points, "PH, code" as "PM, code"; for $p \geq 3$ PH and $\mathsf{PH}_{\text{code}}$ analogous to PM and $\mathsf{PM}_{\text{code}}$.

2 Nodal rotation patch test ($p = 2$)

PM9 passes the constant and the linear strain patch tests on all meshes, as in the paper (Tables 1 and 3); for the quadratic-strain (cubic displacement) shear test, which lies outside Q_2, its results are within 1.7 % of the exact value on all meshes (Table 2). All PP9 and MM9 values for moment and shear coincide with the paper to four digits. The metric functions of the code base (skew coordinates) give the same results for the constant and linear strain tests and comparable ones for the shear test (3.15–3.41). PH ($p = 2$, corner centroid) passes the constant and linear strain tests on all meshes as well, since $P_2(\boldsymbol{x})$ is contained in its trial space; for the shear test it is less accurate than PM9 (3.37–3.61, i.e. 1.8–8.9 % error), because its quadratic complement is parametric.

3 Convergence test (shear loading, $p = 2$)

The displacements of sequence A and of PP9 and MM9 for B and C agree with the paper. PM9 on sequence B yields the values the paper lists for sequence C and vice versa (Section 6). PM9 converges with the optimal rate 2 on the folded meshes of B and C, as reported in the paper, and so do PM in skew coordinates (2.02, 2.13) and PH (2.02, 2.21); PP9 does not (0.70, 2.50 from a negative displacement at 1×1). For $p = 3$ both unsymmetric elements reproduce the field on all meshes (error at round-off level, "–"), whereas PP ($p = 3$) still has energy-norm errors of 9–80 %.

4 Higher degrees

With interior nodes from the bilinear corner map, both unsymmetric elements are Jacobian-unlimited immune for $p = 3, \ldots, 8$ (Table 6): the cubic field is reproduced on all meshes with overlapping and folded elements. The round-off level grows with p and distortion, for PM from 10^{-13} to 10^{-5} ($p = 8$, Meshes c and e), for PH to $7 \cdot 10^{-8}$, in line with the thick-cylinder study. With the interior nodes of the code base, PH is computable in more cases than PM (e.g. $p = 8$, Meshes c and e: $\mathsf{PM}_{\text{code}}$ rejected, $\mathsf{PH}_{\text{code}}$ $e_n \leq 3 \cdot 10^{-11}$). The conventional element has errors of 7–100 % on the distorted meshes for all p.

5 Interior nodes of the code base on these meshes

`pSpectralElement_NodesTopology` places all new nodes isoparametrically, i.e. with the geometry map of the original 8-node (serendipity) element. The serendipity space is contained in Q_p, so the degree-p geometry reproduces the 8-node geometry exactly; the map is also the Gordon–Hall interpolation of the three-node edges. On Meshes b and f, however, this map sends the interior reference point $\boldsymbol{\xi} = (0, 0)$ onto a mid-side node of the same element (e.g. $-\frac{1}{4}\sum \boldsymbol{x}_{\text{corner}} + \frac{1}{2}\sum \boldsymbol{x}_{\text{mid}} = (1, -0.5)$ in the element with corners $(0, -1)$, $(1, -1)$, $(1, 0)$, $(0, 0)$), for every even p. No set of distinct nodes represents this element geometry; the 9th node of the paper (corner centroid) defines a different geometry. Before commit `da84a58` the metric Vandermonde matrix

Table 1: Moment loading, $u_y(A)$ (paper: Table 15; exact 1.5).

	PP9		MM9		PM9 (PM-QUAD9*)		PM, $p=2$		PH, $p=2$	
mesh	present	paper	present	paper	present	paper	skew	code	centr.	code
a (0°)	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000
b (90°)	1.2606	1.2606	1.7935	1.7935	1.5000	1.5000	1.5000	err.	1.5000	err.
c (135°)	1.0372	1.0372	1.6398	1.6398	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000
d (175°)	1.2472	1.2472	1.5285	1.5285	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000
e (225°)	1.0372	1.0372	1.6398	1.6398	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000
f (270°)	1.2606	1.2606	1.7935	1.7935	1.5000	1.5000	1.5000	err.	1.5000	err.

Table 2: Shear loading, $u_y(A)$ (paper: Table 16; exact 3.3125). *: differs from the present result, see Section 6.

	PP9		MM9		PM9 (PM-QUAD9*)		PM, $p=2$		PH, $p=2$	
mesh	present	paper	present	paper	present	paper	skew	code	centr.	code
a (0°)	3.2979	3.2979	3.2979	3.2979	3.2979	3.2979	3.2979	3.2979	3.2979	3.2979
b (90°)	3.0033	1.9088*	3.9559	3.9559	3.2710	3.2710	3.3120	err.	3.3723	err.
c (135°)	-0.4993	-0.4993	3.8294	3.8294	3.3044	3.3044	3.4134	4.1224	3.4016	3.0874
d (175°)	2.5498	2.5498	3.3657	3.3657	3.3677	3.3677	3.1501	3.0344	3.6074	3.0934
e (225°)	-0.4993	-0.4993	3.8294	3.8294	3.3044	3.3044	3.4134	4.1224	3.4016	3.0874
f (270°)	3.0033	1.9088*	3.9559	3.9559	3.2710	3.2710	3.3120	err.	3.3723	err.

Table 3: Tension, $u_x(A)$ (paper: Table 14; exact 1.0). In Tables 1 to 4, "err." marks cases in which the code base stops because interior and edge nodes coincide (Section 5).

	PP9		MM9		PM9 (PM-QUAD9*)		PM, $p=2$		PH, $p=2$	
mesh	present	paper	present	paper	present	paper	skew	code	centr.	code
a (0°)	1.0000	1.0	1.0000	1.0	1.0000	1.0	1.0000	1.0000	1.0000	1.0000
b (90°)	1.0000	1.0	1.1300	1.0*	1.0000	1.0	1.0000	err.	1.0000	err.
c (135°)	1.0000	1.0	1.4633	1.0*	1.0000	1.0	1.0000	1.0000	1.0000	1.0000
d (175°)	1.0000	1.0	1.0285	1.0*	1.0000	1.0	1.0000	1.0000	1.0000	1.0000
e (225°)	1.0000	1.0	1.4633	1.0*	1.0000	1.0	1.0000	1.0000	1.0000	1.0000
f (270°)	1.0000	1.0	1.1300	1.0*	1.0000	1.0	1.0000	err.	1.0000	err.

was then singular (cond $\approx 10^{18}$–10^{20}) and the computed displacements were silently zero. Since then the code base keeps the isoparametric placement but stops with an error: `pSpectralElement_NodesTopology` on coincident nodes, and `ShapeFunctions2D_Metric` when its interpolation matrix is singular to working precision (rcond $< \varepsilon$), which also covers the nearly non-unisolvent node sets of Meshes c–e for high p, where the accuracy had been destroyed (Table 6). The "code" columns and $\mathsf{PM}_{\text{code}}$, $\mathsf{PH}_{\text{code}}$ rows show these cases as "err.". With the Lagrange basis of the metric functions (bounding box of the skew coordinates) the interpolation matrix is better conditioned than with monomials: five $\mathsf{PM}_{\text{code}}$ cases that the check rejected with the monomial basis ($p = 3$, Mesh d; $p = 6, 7$, Meshes c and e) are now computed and pass the patch test ($e_n \leq 4 \cdot 10^{-7}$), and the round-off-dominated values change towards the exact ones (e.g. $p = 2$, Mesh d, moment: $u_y(A) = 1.49957 \to 1.50000$). On all meshes of the test suite, the Lee & Bathe input files and the thick-cylinder meshes with $\alpha \leq 30°$ ($p \leq 8$) neither check is triggered. The study results with interior nodes from the bilinear corner map (PM, $p \geq 3$) show that such meshes can be computed if a different geometry is acceptable.

6 Inconsistencies in the paper

1. $E = 1500$ is stated, the exact values correspond to $E = 2000$ (Section 1).
2. Fig. 8, moment loading: nodal forces at nodes 8/9 and 14/13 interchanged (Section 1).
3. Table 16, PP-QUAD9, Meshes b and f: 1.9088; the present value 3.0033 equals the paper's own PP-QUAD9 value for the same mesh in Table 18 (1×1 super-element of sequence B).
4. Tables 18 and 19, PM-QUAD9* columns: interchanged. The present values for sequence B (3.2710,

Table 4: $u_y(A)$ for the sequences A, B, C (paper: Tables 17–19; exact 3.3125).

		PP9		MM9		PM9		PM, $p=2$		PH, $p=2$	
seq.	mesh	present	paper	present	paper	present	paper	skew	code	centr.	code
A	1×1	3.2979	3.2979	3.2979	3.2979	3.2979	3.2979	3.2979	3.2979	3.2979	3.2979
	2×2	3.3106	3.3106	3.3106	3.3106	3.3106	3.3106	3.3106	3.3106	3.3106	3.3106
	4×4	3.3123	3.3123	3.3123	3.3123	3.3123	3.3123	3.3123	3.3123	3.3123	3.3123
B	1×1	3.0033	3.0033	3.9559	3.9559	3.2710	3.3044*	3.3120	err.	3.3723	err.
	2×2	3.1768	3.1768	3.9519	3.9519	3.3000	3.3057*	3.3185	err.	3.3229	err.
	4×4	3.2662	3.2661	3.9598	3.9598	3.3110	3.3116*	3.3121	err.	3.3149	err.
C	1×1	-0.4993	-0.4993	3.8294	3.8294	3.3044	3.2710*	3.4134	4.1224	3.4016	3.0874
	2×2	2.9218	2.9218	3.9208	3.9208	3.3057	3.3001*	3.2600	3.2935	3.3218	3.2938
	4×4	3.1531	3.1531	3.9660	3.9660	3.3116	3.3110*	3.3067	3.3105	3.3121	3.3075

Table 5: Least-squares rates of the energy-norm strain error, Eq. (24) of the paper (paper: Fig. 15). PP on folded meshes: the parametric strain is singular where $\det \boldsymbol{J} = 0$, so the energy error of PP is not well defined for B and C (here: signed $\det \boldsymbol{J}$). "–": error at round-off level (field reproduced).

	PP9		PM9		PM, $p=2$		PH, $p=2$		$p=3$		
seq.	present	paper	present	paper	skew	code	centr.	code	PP	PM	PH
A	1.99	1.93	1.99	1.93	1.99	1.99	1.99	1.99	–	–	–
B	0.70	1.05	2.00	2.11	2.02	err.	2.02	err.	0.64	–	–
C	2.50	0.85	1.98	1.97	2.13	2.41	2.21	1.95	0.94	–	–

3.3000, 3.3110) are the paper's Table 19 entries (3.2710, 3.3001, 3.3110), those for C the Table 18 entries; the 1×1 values of Table 16 (b: 3.2710, c: 3.3044) confirm the present assignment.

5. Table 14, MM-QUAD9: 1.0 for all meshes. With a stiffness matrix that reproduces all MM9 moment and shear values to four digits, MM9 gives 1.1300 (b, f), 1.4633 (c, e), 1.0285 (d), also with load vectors computed from the metric functions (Eq. (22) of the paper). MM9 does not pass the constant strain patch test on these meshes; the paper's entries could not be reproduced.

Apart from these items, every 9-node value of the paper for $\theta \geq 90°$ is reproduced to four digits. The 8-node results (Tables 8–13) are not reproduced: the code base provides only tensor-product (Lagrange) elements.

7 Reproduction

`mesh/gen_panel_meshes.py` (gmsh), `matlab/rp_run.m` (all computations, GNU Octave 8.4 or MATLAB; code base after commit `e293da1`), `postprocess.py` (tables). Paper values: `results/paper_values.csv`, `results/paper_rates.csv`.

References

[1] S. Rajendran. A technique to develop mesh-distortion immune finite elements. *Computer Methods in Applied Mechanics and Engineering*, 199:1044–1063, 2010. doi: 10.1016/j.cma.2009.11.017.

Table 6: Shear loading on Meshes a–f: relative nodal error e_n for $p = 3, \ldots, 8$ (cubic field, contained in the PM trial space for $p \geq 3$). PM: interior nodes from the bilinear corner map; $\mathsf{PM}_{\text{code}}$: interior nodes of `pSpectralElement_NodesTopology`; "err.": the code base stops (coincident nodes or singular metric interpolation).

	a (0°)	b (90°)	c (135°)	d (175°)	e (225°)	f (270°)
PP, $p = 3$	1e-14	2e-1	1e-1	2e-1	1e-1	2e-1
PM, $p = 3$	2e-14	2e-14	7e-14	1e-13	2e-14	3e-14
$\mathsf{PM}_{\text{code}}$, $p = 3$	7e-14	7e-13	1e-12	7e-8	6e-12	1e-13
PH, $p = 3$	3e-14	3e-14	3e-13	2e-13	2e-13	6e-13
$\mathsf{PH}_{\text{code}}$, $p = 3$	2e-14	5e-14	9e-14	6e-8	8e-14	1e-14
PP, $p = 4$	1e-13	8e-2	5e-1	2e-1	5e-1	8e-2
PM, $p = 4$	1e-13	2e-13	3e-12	3e-11	3e-12	2e-13
$\mathsf{PM}_{\text{code}}$, $p = 4$	1e-13	err.	3e-11	err.	5e-10	err.
PH, $p = 4$	1e-13	2e-14	6e-12	5e-13	1e-11	1e-14
$\mathsf{PH}_{\text{code}}$, $p = 4$	1e-13	err.	1e-12	err.	1e-12	err.
PP, $p = 5$	8e-13	7e-2	9e-2	2e-1	9e-2	7e-2
PM, $p = 5$	9e-13	1e-10	6e-13	8e-11	6e-13	3e-11
$\mathsf{PM}_{\text{code}}$, $p = 5$	8e-13	4e-10	8e-10	err.	6e-10	5e-9
PH, $p = 5$	9e-13	8e-13	2e-12	2e-11	5e-12	2e-13
$\mathsf{PH}_{\text{code}}$, $p = 5$	6e-13	7e-13	2e-12	err.	2e-12	2e-13
PP, $p = 6$	7e-13	9e-2	2e-1	1e-1	2e-1	9e-2
PM, $p = 6$	6e-13	1e-11	6e-11	3e-11	2e-11	1e-11
$\mathsf{PM}_{\text{code}}$, $p = 6$	6e-13	err.	1e-8	err.	7e-9	err.
PH, $p = 6$	5e-13	2e-13	2e-11	1e-10	1e-11	2e-13
$\mathsf{PH}_{\text{code}}$, $p = 6$	4e-13	err.	2e-12	err.	2e-12	err.
PP, $p = 7$	3e-13	1e-1	9e-2	1e-1	9e-2	1e-1
PM, $p = 7$	4e-14	2e-11	4e-10	8e-10	1e-9	4e-11
$\mathsf{PM}_{\text{code}}$, $p = 7$	3e-13	2e-6	8e-8	err.	4e-7	2e-6
PH, $p = 7$	5e-13	5e-12	1e-10	3e-10	2e-11	9e-12
$\mathsf{PH}_{\text{code}}$, $p = 7$	1e-12	2e-12	4e-12	err.	3e-11	2e-12
PP, $p = 8$	3e-12	1e-1	1e0	1e-1	1e0	1e-1
PM, $p = 8$	3e-12	3e-9	8e-6	1e-7	1e-5	5e-9
$\mathsf{PM}_{\text{code}}$, $p = 8$	3e-12	err.	err.	err.	err.	err.
PH, $p = 8$	3e-12	1e-10	7e-8	6e-9	1e-8	4e-11
$\mathsf{PH}_{\text{code}}$, $p = 8$	2e-12	err.	1e-11	err.	3e-11	err.